Hybrid Depth Optimization and Multi-Representation Geometric-Semantic Fusion For Fast Occupancy Prediction

Anonymous ICCV submission

Paper ID 10843

Abstract

3D occupancy prediction is an emerging and challenging perception task in autonomous driving, which aims to model the scene using 3D voxel grids and perform spatial occupancy inference by combining both geometric and semantic information. Existing vision-based occupancy prediction methods fail to fully leverage 3D features due to the lack of depth information, and while the LSS pipeline estimates depth via a uniform distribution, it remains unable effectively capture the 3D geometric details of objects. Therefore, we propose HGS-OCC, a fast occupancy prediction method that combines hybrid depth optimization for 3D modeling and fuses 3D geometry with 2D semantics through multi-representation integration. Hybrid depth optimization generates relative depth maps using Depth Angthing V2, which are then refined with LiDAR's metric depth to provide accurate 3D geometric details. The multi-representation geometric-semantic fusion combines 3D geometric with 2D semantic information to merge point, voxel, and BEV features to significantly enhance the representation of geometric information. Experiments on the nuScenes dataset demonstrate the effectiveness of our method and offer a new perspective for vision-based occupancy prediction. The code will be released at `https://github.com`

1. Introduction

Accurate 3D perception of the surrounding environment forms the cornerstone of modern autonomous driving systems and robotics, ensuring efficient planning and safe control [5, 9]. In recent years, advancements in 3D object detection [17, 20, 23, 41, 45] and semantic segmentation [11, 13, 40, 42, 50] have significantly propelled the field of 3D perception. However, object detection relies on strict bounding boxes, making it difficult to recognize arbitrary shapes or unknown objects, while semantic segmentation struggles with fine-grained classification in complex scenes, especially under occlusion and overlap. In this context, 3D

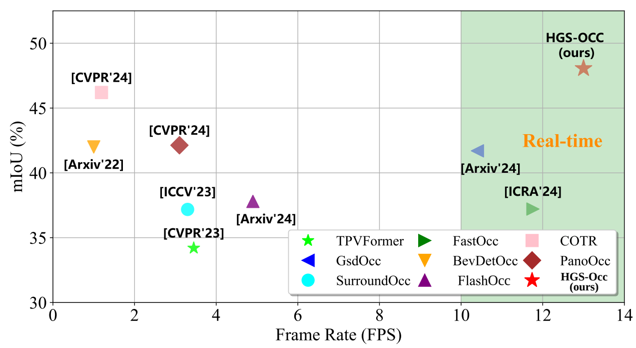
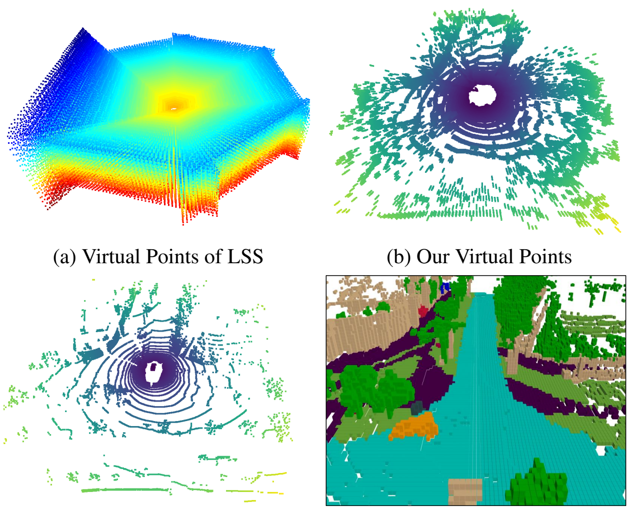


Figure 1. Comparisons of the mIoU and inference speed (FPS) of various 3D occupancy prediction methods on the Occ3D-nuScenes validation set. HGSOCC achieves higher accuracy and real-time inference speed.

semantic occupancy prediction [32, 33] offers a more comprehensive approach to environment modeling. It simultaneously estimates the geometric structure and semantic categories of scene voxels, assigns labels to each 3D voxel, and provides a more complete perception, showing stronger robustness to arbitrary shapes and dynamic occlusions.

Despite the significant advantages of 3D occupancy prediction in tasks such as object detection and semantic segmentation, existing methods [1, 14, 16, 17, 48] often face challenges related to slow inference speed (see Fig.1) (e.g., only 1-3 FPS on Nvidia 4090 [24]) and high memory usage (e.g., >10,000 MB [24, 36]) due to the high computational cost of 3D voxel features. While LSS-based methods have somewhat improved inference speed, they rely on evenly spaced depth distribution estimation, i.e., virtual point seeds uniformly distributed in 3D space, as shown in Fig.2(a). This approach struggles to effectively leverage 3D geometric structure information, resulting in difficulties synchronizing improvements in both performance and speed.

To address these challenges, we propose a fast 3D occupancy prediction, which integrates hybrid depth optimization modeling and fuses 3D geometric information with 2D semantic information through multi-representation. Specifically, we decompose the 3D occupancy prediction task into



(a) Virtual Points of LSS (b) Our Virtual Points

(c) Original Point Cloud (d) Occupancy grid

Figure 2. (a) Visualization of virtual point seeds generated by the LSS pipeline, uniformly distributed with low 3D geometric efficiency. (b) Visualization of our virtual points, using hybrid depth optimization to preserve 3D structure and expand scene coverage. (c) Raw point cloud visualization. (d) The corresponding occupancy grid.

two steps: virtual point seed generation and feature aggregation. The goal of virtual point seed generation is to create useful virtual point seeds around co-visible points and targets detectable in images but not captured by LiDAR (e.g., black vehicles), thus preserving the 3D geometric information and extending coverage across the entire scene. First, after extracting image features, we apply hybrid depth optimization modeling, generating dense relative depth maps with the fine-tuned Depth Anything V2 model and optimizing them using sparse metric depth maps provided by LiDAR. Finally, we construct multi-level projection depth maps using bidirectional linear discretization and back-project them to generate virtual point seeds.

Second, after generating virtual point seeds that retain both image pixel features and 3D geometric structure, we introduce a multi-representation geometric-semantic feature aggregation module, designed to efficiently fuse 3D geometry and 2D semantic features using submanifold 3D sparse convolution. By extracting 3D sparse features of virtual point seeds through an improved point-voxel network, we obtain multi-level point features, BEV features, and voxel features. The fusion of 3D geometry and 2D semantic information across dimensions such as point, BEV, and voxel significantly improves the accuracy of occupancy prediction. The experiments on the Occ3d-nuscenes and Surround-nuscenes datasets demonstrate the effectiveness of our method and offer a new perspective for vision-based occupancy prediction. Our contributions can be summarized as follows:

- We introduce a fast occupancy prediction framework, termed HGS-OCC, which aims to combine hybrid depth optimization for 3D modeling, and achieves a balance between speed and accuracy by fusing 3D geometry and 2D semantic information through multi-representations.
- We propose a novel virtual point generation method that leverages hybrid depth optimization, utilizing the dense relative depth maps generated by fine-tuning the Depth Anything model in conjunction with metric depth optimization, preserving the 3D geometric structure of objects while covering the entire scene.
- We introduce a multi-representation geometric-semantic fusion module for 3D geometry and 2D semantics, which employs 3D sub-manifold sparse convolutions to generate multi-scale sparse features and fuses 3D geometric structure and 2D semantic features across point, BEV, and voxel spaces.
- Our method achieves a balance between performance and speed on the nuScenes dataset, demonstrating the effectiveness of our approach.

2. Related Work

2.1. Vision-based BEV Perception

In recent years, vision-based Bird's-Eye View (BEV) perception has made significant advancements and has emerged as a critical component in autonomous driving due to its high computational efficiency, cost-effectiveness, stability, and versatility [12]. By transforming 2D image features into a unified and comprehensive 3D BEV representation through view transformation, various tasks, including 3D object detection and map segmentation, have been consolidated into a unified framework. View transformation can be broadly categorized into two types: one relies on explicit depth estimation to form a pseudo point cloud and construct the 3D space [6, 15, 19, 29, 30], while the other pre-defines the BEV space and implicitly models depth information by mapping image features to corresponding 3D positions through spatial cross-attention[10, 34, 35, 43, 49]. Although BEV perception excels in 3D object detection, it still faces challenges in handling corner cases in driving scenarios, such as irregular obstacles and out-of-vocabulary objects, which are critical for ensuring the safety of autonomous driving. To address these challenges, the 3D occupancy prediction task has been proposed, rapidly emerging as a promising solution in autonomous driving.

2.2. 3D Occupancy Prediction

The 3D occupancy prediction task has received widespread attention due to its rich geometric information and its significant advantages over 3D object detection in general object recognition. MonoScene [1] is a pioneering work that uses only RGB input. TPVFormer [8] integrates multi-camera inputs and leverages transformer-based techniques

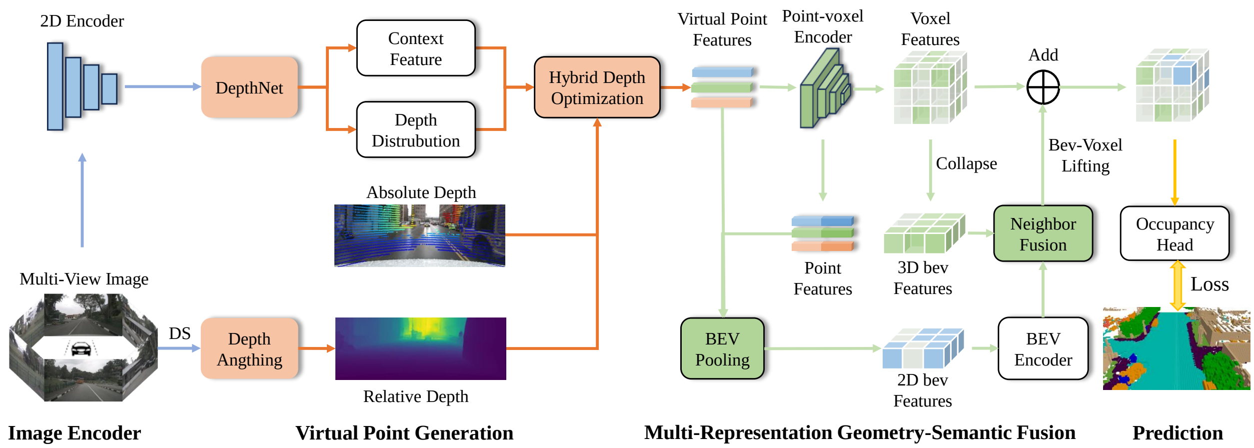


Figure 3. The overall architecture of HGS-OCC. T-frame multi-view images are fed into the image backbone for feature extraction while simultaneously downsampling for depth estimation via Depth Angthing. The image features and depth maps are refined through a hybrid depth optimization module to generate virtual point seeds with image semantic features. These seeds are passed through a point-voxel encoder to produce point, BEV, and voxel features, which are fused with the original virtual point features through multi-representation, yielding voxel features for supervised learning.

to enhance features into a tri-perspective space. SurroundOcc [38] extends the high-dimensional BEV features into occupancy-related information, applying spatial cross-attention for geometric generation. VoxFormer [16] introduces a two-stage transformer framework for semantic scene completion, enabling the output of full 3D voxel semantics from 2D images. FlashOcc [46] optimizes the process by converting channels into height, thus elevating BEV outputs into 3D space, which improves operational efficiency. FBOcc [18] presents a novel forward-backward view transformation module, tackling the limitations of different view transformations. While methods like UniOcc [28] and RenderOcc [27] use Neural Radiance Fields [37] for direct 3D semantic occupancy predictions, their rendering speed limits efficiency. FastOcc [4] improves the occupancy prediction head to accelerate inference speed. Lastly, COTR [24] uses explicit-implicit view transformation combined with coarse-to-fine semantic grouping to build a compact 3D occupancy representation. They utilize LSS to generate uniformly distributed virtual point seeds, without considering the sparsity of the scene and the insufficient utilization of 3D geometric structure information.

3. Methods

3.1. Problem Formulation

In this work, we aim to predict 3D occupancy states and semantic classifications of the surrounding environment using multi-view images. The input consists of T consecutive frames from N surround-view cameras, represented as $X \in \mathbb{R}^{N \times H_C \times W_C \times 3}$, where H_C and W_C are the image height and width, respectively. A neural network is trained to generate a 3D occupancy voxel map $Y \in \mathbb{R}^{H \times W \times D \times L}$. H, W, and D represent the scene volume dimensions, and each voxel in Y is labeled as either unknown, occupied, or assigned a semantic category from $\{0 \text{ to } L\}$, with L being the number of categories and 0 indicating unoccupied grids.

3.2. Overview

An overview of HGS-OCC is shown in Fig.3. It mainly consists of four key modules: image feature encoder to extract image features, hybrid depth optimization to generate virtual point seed, multi-representation geometric-semantic fusion module for fusing 3D geometry and 2D semantics features and the occupancy prediction head for final output.

3.3. Image Encoder

The image encoder is designed to capture features across multiple views, forming the basis for 2D-3D view transformation. Given the RGB images from surround-view cameras, we begin by applying a pre-trained backbone network, such as the well-known ResNet [3], to extract multi-layer image features $F_C \in \mathbb{R}^{N_C \times C \times H \times W}$. These features are then aggregated via a Feature Pyramid Network (FPN) [21], which merges both fine and coarse features before downsampling them to a designated scale, typically 1/16.

3.4. Virtual Point Generation Based Hybrid Depth Optimization

In 3D perception tasks, the LSS pipeline is widely employed to map image features into BEV representations by generating virtual points for each pixel within a predefined depth range. It predicts depth distribution weights α and context features c, with the feature representation at depth d defined as $p_d = \alpha_d c$, resulting in 3D virtual points enriched with image features. However, the uniform distribution of virtual points in 3D space results in inefficient feature utilization, as occupied regions are sparse in the occupancy

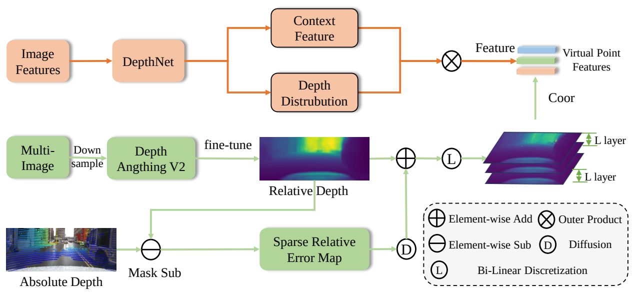


Figure 4. The overall framework of the hybrid depth optimization-based virtual point generation. This method combines dense depth maps from Depth Anything V2 with sparse metric depth maps, using co-visible neighborhood diffusion for local calibration. To mitigate depth uncertainty, discretization techniques are applied to generate multi-layer depth maps, which are then projected onto virtual point seeds enriched with image semantic features, effectively preserving the 3D structure of objects.

grid, limiting the method's ability to capture the 3D geometric structure effectively.

Therefore, we propose a novel hybrid depth optimization for virtual point generation that preserves 3D structural features by integrating multi-source depth information and discrete spatial sampling strategies, as shown in Fig.4. First, Depth Anything V2 is used to generate a dense depth map with high relative accuracy but significant metric depth bias, which is then refined using sparse metric depth maps. This refinement leverages a co-visible neighborhood with a tunable radius ($r = 1$), where metric depth values serve as anchor points for local calibration, improving geometric accuracy while preserving local depth distribution. To handle depth uncertainty, we employ linear-increasing [30] and linear-decreasing discretization to create multi-layer discrete depth maps, ensuring scene coverage. These maps are subsequently projected to getting virtual point seeds, effectively preserving the 3D structure of the object.

Given the dense relative depth map $D_r^{dense}(p)$ and sparse metric depth map $D_m^{sparse}(p)$, the sparse relative error map $E^{sparse}(p)$ is computed by performing a masked differencing operation in the co-visible regions $\mathcal{P}$, while the remaining regions are set to zero. This process is expressed as:

$$E^{sparse}(p) = \begin{cases} \left|D_r^{dense}(p) - D_a^{sparse}(p)\right| & \text{if } p \in \mathcal{P} \\ 0 & \text{otherwise} \end{cases}, \tag{1}$$

where $\mathcal{P}$ denotes the co-visible region, $D_r^{dense}(p)$ is the depth value from the dense relative depth map, and $D_a^{sparse}(p)$ is the corresponding depth value from the sparse metric depth map.

To propagate the influence of metric depth constraints, we diffuse depth values to neighboring pixels within a certain radius. Specifically, for each pixel p, its depth value $D(p)$ is assigned to all neighboring pixels within a predefined radius r, as follows:

$$D^{diff}(q) = D(p) \quad \text{if} \quad \|p - q\| \leq r, \tag{2}$$

where $N(p)$ represents the neighboring pixels within the radius r, and $\|p - q\|$ is the distance between pixels p and q. In the overlapping region, the average depth is computed while keeping the depth of the current pixel unchanged, thus generating a more accurate dense relative error map.

The final enhanced depth map $D^{enhanced}(p)$ is obtained by element-wise addition of the dense relative depth map $D_r^{dense}(p)$ and the diffused relative error map $E^{dense}(p)$:

$$D^{enhanced}(p) = D_r^{dense}(p) + E^{dense}(p). \tag{3}$$

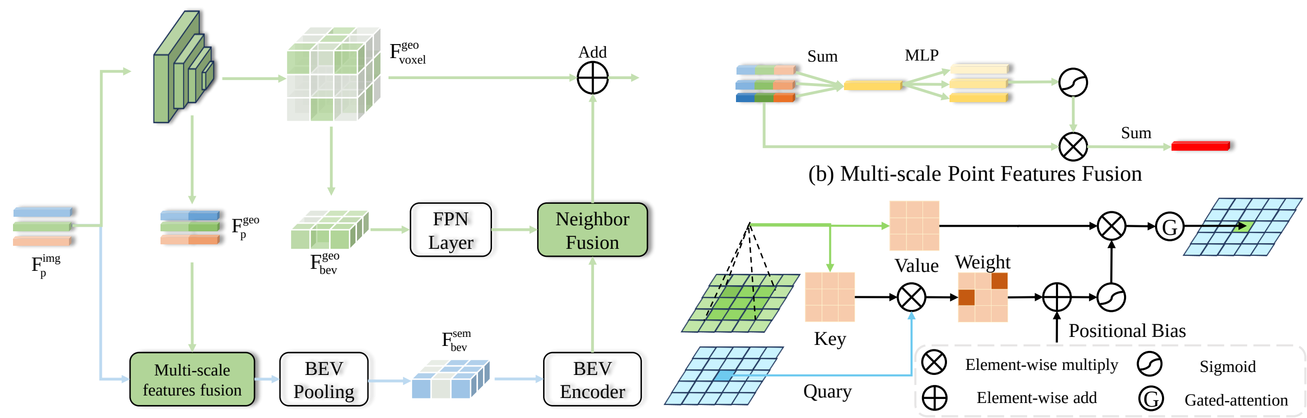
For regions where LiDAR fails to scan or surfaces with low reflectivity (e.g., black vehicles), depth values are compensated using dense depth estimation. Furthermore, due to the lower accuracy of depth estimation on low-resolution images, performance degrades significantly with high upsampling ratios in the LSS pipeline. Therefore, we fine-tune the metric depth at the corresponding feature scales using Depth Anything V2, incorporating a pixel reprojection error constraint to mitigate such errors.

Due to the projection deviation from 2D pixels to 3D points, we apply bidirectional linear discretization to the extended depth map, performing linear increase and decrease on non-zero values within a certain range to obtain multi-layer depth maps, which are then back-projected to generate 3D virtual point seed coordinates. Finally, the image textual features F_t and depth distribution weights D_w are computed using the outer product $F_t \otimes D_w$, providing image features that collaboratively construct virtual point seeds with image features F_p^{img}.

3.5. Multi-Representation Geometric-Semantic Fusion

Existing vision-based perception methods typically generate virtual point seeds via LSS pipeline and rely on voxel or BEV features for 3D feature encoder. However, the feature sparsity caused by uniform sampling makes traditional sparse voxel encoding ineffective in capturing the geometric structure of 3D objects while increasing computational burden. Although BEV features reduce computational complexity, they are limited by the lack of height information, thus restricting model performance.

Therefore, we propose a multi-representation geometric-semantic fusion method, as shown in Fig.5, that enhances feature representation by preserving the 3D geometric structure of virtual point seeds while collaboratively fusing hierarchical features in point clouds, BEV, and voxel spaces. Specifically, we first construct a point-voxel encoder based on sparse subconv3D [31], using virtual point features F_p^{img} with image semantic as priors to extract multi-scale point



(a) Multi-Representation Geometry-Semantic Fusion (c) BEV Neighbor Feature Fusion

Figure 5. (a) The overall architecture of Multi-representation geometric-semantic fusion, where a point-voxel encoder based on sparse subconv3D fuses features across point, BEV, and voxel representations to enhance feature representation. (b) Multi-scale geometric point features are fused with image-semantic enriched virtual point features to construct semantic BEV point features. (c) Semantic features are queried in the local neighborhood of geometric features and fused via pixel-wise attention, with final enhancement through a gated attention mechanism.

features F_p^{geo}, geometric BEV features F_{bev}^{geo}, and geometric voxel features F_{voxel}^{geo}. The multi-scale point features are fused with the original virtual point features through attention-guided scale selection. The constructed semantic BEV features are enhanced by neighborhood fusion with geometric BEV features. Finally, the enhanced geometric-semantic BEV features are lifted to voxel features and fused with geometric voxel features for supervised learning.

After obtaining multi-scale point features, traditional fusion methods typically use tensor concatenation or summation, treating features from different scales equally. However, these features have distinct geometric priors and focus on different aspects of scene understanding. Therefore, we apply a scale-wise weighting strategy to multi-scale point features and the original virtual point features with image semantics, reallocating the importance of each scale to construct geometry-semantic point features.

First, the features from multiple scales are summed to obtain the combined feature:

$$\hat{O} = \sum_{k=1}^{K} O_k, \tag{4}$$

where O_k represents the feature from the k-th scale, and K is the total number of scales.Then, using the computed attention weights, the final fused feature $F_p^{\text{fuse}}$is obtained by weighting and summing all the features:

$$F_p^{\text{fuse}} = \sum_{k=1}^{K} \left(\sigma(\text{MLP}(\hat{O})) \cdot O_k \right), \tag{5}$$

where $\sigma(\text{MLP}(\hat{O}))$ represents the attention weight, σ denotes the sigmoid function and O_k is the feature from the k-th scale.

After obtaining the fused point features F_p^{fuse}, BEV pooling is performed to obtain 2D semantic BEV features F_{bev}^{sem}. For multi-scale geometric BEV features, we fuse them through a neck module to a specific downsampled scale feature F_{bev}^{geo}. We then adopt the neighborhood attention from AAA to extract local patch features corresponding to pixels from the F_{bev}^{sem}, and dynamically adjust the weights through gated attention to selectively enhance the geometry-semantic BEV features.

Specifically, the image semantic features F_{bev}^{sem} are compressed into a sequence of feature vectors and projected through a linear layer to obtain the query features $Q \in \mathbb{R}^{n \times q}$. Similarly, the geometric features F_{bev}^{geo} are projected to obtain the key $K \in \mathbb{R}^{n \times q}$ and value $V \in \mathbb{R}^{n \times v}$ features. The semantic features query within the local neighborhood of the geometric features, performing pixel-wise local attention feature fusion, they are computed is as follows:

$$F_{\text{nbr}} = \sigma \left(\frac{Q^i \cdot (K^{n(i)})^T + B(i, n(i))}{\sqrt{v}} \right) \cdot V^i, \tag{6}$$

where $n(i)$ represents the neighborhood of size k centeredat the query point i, $B(i, n(i))$ represents the relative position bias, and σ is the Softmax function. Finally, through a gated attention mechanism, the fused feature $F_{bev}^{\text{geo-sem}}$ is obtained as:

$$F_{bev}^{\text{geo-sem}} = \sigma(\text{Conv}(f_{\text{Avg}}(F_{\text{nbr}}))) \cdot F_{\text{nbr}}, \tag{7}$$

where σ denotes the Sigmoid function, Conv represents a linear transformation matrix (e.g., 1×1 convolution), and f_{Avg} denotes adaptive average pooling.

Finally, we introduce a channel-to-height transformation module to convert the geometric-semantic features $F_{bev}^{\text{geo-sem}}$ into voxel features. This module reshapes the BEV features from $F_{bev}^{\text{geo-sem}} \in \mathbb{R}^{B \times C \times H \times W}$ to $F_{voxel}^{\text{geo-sem}} \in$

ICCV #10843

Method	Input	Backbone	Visible Mask	mIoU	others	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. suf.	other flat	sidewalk	terrain	manmade	vegetation	Time(ms)
TPVFormer [8]	C	R-50	✔	34.2	7.68	44.01	17.66	40.88	46.98	15.06	20.54	24.69	24.66	24.26	29.28	79.27	40.65	48.49	49.44	32.63	29.82	289.85
OccFormer [48]	C	R-50	✔	37.4	9.15	45.84	18.20	42.80	50.27	24.00	20.80	22.86	20.98	31.94	38.13	80.13	38.24	50.83	54.3	46.41	40.15	-
FBOcc [18]	C	R-50	✔	42.1	14.30	49.71	30.0	46.62	51.54	29.3	29.13	29.35	30.48	34.97	39.36	83.07	47.16	55.62	59.88	44.89	39.58	-
PanoOcc [36]	C	R-101	-	42.13	11.67	50.48	29.64	49.44	55.52	23.29	33.26	30.55	30.99	34.43	42.57	83.31	44.23	54.40	56.04	45.94	40.40	322.58
FastOcc [4]	C	R-101	✔	40.75	12.86	46.58	29.93	46.07	54.09	23.74	31.10	30.68	28.52	33.08	39.69	83.33	44.65	53.90	55.46	42.61	36.50	221.2
SurroundOcc [38]	C	R-101	✔	37.1	8.97	46.33	17.08	46.54	52.01	20.05	21.47	23.52	18.67	31.51	37.56	81.91	41.64	50.76	53.93	42.91	37.16	303.03
BEVDet4D [6]	C	Swin-B	✔	42.5	12.37	50.15	26.97	51.86	54.65	28.38	28.96	29.02	28.28	37.05	42.52	82.55	43.15	54.87	58.33	48.78	43.79	1000.0
FlashOcc [46]	C	Swin-B	✔	43.52	13.31	51.62	28.07	50.91	55.69	27.46	31.05	29.98	29.20	38.86	43.68	83.87	45.63	56.33	59.01	50.63	44.56	909.1
COTR [24]	C	Swin-B	✔	46.2	**14.85**	53.25	**35.19**	50.83	57.25	35.36	34.06	33.54	**37.14**	38.99	44.97	84.46	**48.73**	57.60	61.08	51.61	46.72	840.34
HyDRa [39]	C+R	R-50	-	44.40	-	-	-	52.3	56.3 -	-	35.9	35.10	-	-	44.1	-	-	-	-	-	-	-
OCCFusion [25]	C+L	R-101	-	46.79	11.65	47.81	32.07	57.27	57.51	31.80	**40.11**	47.35	33.74	45.81	50.35	78.79	37.17	44.36	53.36	63.18	63.20	-
HGS-OCC	C	R-50	✔	**48.34**	11.57	**56.32**	26.35	56.89	**60.33**	32.57	34.23	46.49	32.28	**47.62**	48.97	83.77	47.45	**58.00**	**61.61**	**70.67**	**67.65**	**75.7**

Table 1. 3D Occupancy prediction performance on the Occ3D-nuScenes dataset. We present the mean IoU over categories and the IoUs for different classes. The best scores for each class are highlighted in bold. In the Input, the C, L, and R denote camera, LiDAR, and radar, respectively. In the backbone, R represents ResNet, while Swin stands for Swin Transformer.

Method	RayIoU	Input	Input Size	Backbone	RayIoU_{1m}	RayIoU_{2m}	RayIoU_{4m}	mIoU
RenderOcc [27]	19.5	C	512 × 1408	Swin-B	13.4	19.6	25.5	24.4
SimpleOcc [2]	22.5	C	336 × 672	R101	17.0	22.7	27.9	31.8
BEVDetOcc(8f) [7]	32.6	C	384 × 704	R50	26.6	33.1	38.2	39.3
BEVFormer(4f) [17]	32.4	C	900 × 1600	R101	26.1	32.9	38.0	39.2
FB-Occ(16f) [18]	33.5	C	256 × 704	R50	26.7	34.1	39.7	39.1
SparseOcc(16f) [22]	36.1	C	256 × 704	R50	30.2	36.8	41.2	30.9
Panoptic-FlashOcc [47]	35.2	C	256 × 704	R50	29.4	36.0	40.1	29.4
Panoptic-FlashOcc(8f) [47]	38.5	C	256 × 704	R50	32.8	39.3	43.4	31.6
HGS-OCC	45.4	C	256 × 704	R50	42.0	45.8	48.4	41.3

Table 2. 3D Occupancy Prediction Performance on Occ3D-nuScenes Dataset Without Visual Masks. (xf) means use x frames for temporal fusion.

$\mathbb{R}^{B\times C_N\times D\times H\times W}$, where B, C, W, H, and D denote the batch size, channel number, class number, and the dimensions of the 3D space, respectively, with $C = C_N \times D$. Then, the transformed features are added to the geometric voxel features F^{geo}_{voxel} to obtain the final voxel features, which are used for supervised learning.

3.6. Occupancy Prediction

After obtaining the voxel features through multi-representation fusion, the occupancy prediction head maps the voxel features $Fvoxel^{\text{geo-sem}}$ to $Ffinal \in \mathbb{R}^{B\times L\times D\times H\times W}$ using a 3×3×3 convolution followed by two linear layers. Here, B, L, W, H, and D represent the batch size, number of categories, and the dimensions of the 3D space, respectively, with each dimension contributing to the final occupancy map prediction.

4. Experiments

We conduct experiments on the large-scale benchmark dataset Occ3D-nuScenes and SurroundOcc-nuScenes to validate the efficacy of our proposed methods. Additionally, we conduct ablation experiments to verify the effectiveness of each component in our method.

4.1. Datasets

Occ3D-nuScenes [32] is a large-scale autonomous driving dataset containing 1,000 urban scenes under various conditions, split into 700 training, 150 validation, and 150 test scenes. The occupancy grid spans -40m to 40m on the X and Y axes and -1m to 5.4m on the Z axis, with a voxel size of 0.4m × 0.4m × 0.4m. It includes 17 semantic labels, comprising 16 object classes and an additional 'empty' class.

Additionally, the Surround-nuScenes dataset extends the occupancy grid prediction range, with the X and Y axes spanning from -50m to 50m and the Z axis from -5m to 3m. The grid resolution is 0.5m × 0.5m × 0.5m, providing a larger and more detailed spatial coverage for modeling complex environments in autonomous driving.

4.2. Implementation Details

We use ResNet-50 as the default image backbone and the extended SPVCNN [31] encoder as the point-voxel encoder. The model is trained on a GeForce RTX 4090 GPU using the AdamW optimizer with a learning rate of 1e-4 and gradient clipping. In virtual point generation , the bilinear incremental discretization range and the number of diffusion feature layers are set to 1 m and 4, respectively. To enhance performance, an exponential moving average (EMA) hook is adopted. Test-time augmentation techniques are not em-

ICCV
#10843

Method	Input	Backbone	mIoU	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. suf.	other flat	sidewalk	terrain	manmade	vegetation
SurroundOcc [38]	C	R-101	20.3	20.5	11.6	28.1	30.8	10.7	15.1	14.0	12.0	14.3	22.2	37.2	23.7	24.4	22.7	14.8	21.8
OccFormer [48]	C	R-101	20.1	21.1	11.3	28.2	30.3	10.6	15.7	14.4	11.2	14.0	22.6	37.3	22.4	24.9	23.5	15.2	21.1
C-CONet [33]	C	R-101	18.4	18.6	10.0	26.4	27.4	8.6	15.7	13.3	9.7	10.9	20.2	33.0	20.7	21.4	21.8	14.7	21.3
FB-Occ	C	R-101	19.6	20.6	11.3	26.9	29.8	10.4	13.6	13.7	11.4	11.5	20.6	38.2	21.5	24.6	22.7	14.8	21.6
RenderOcc	C	R-101	19.0	19.7	11.2	28.1	28.2	9.8	14.7	11.8	11.9	13.1	20.1	33.2	21.3	22.6	22.3	15.3	20.9
L-CONet [33]	L	-	17.7	19.2	4.0	15.1	26.9	6.2	3.8	6.8	6.0	14.1	13.1	39.7	19.1	24.0	23.9	25.1	35.7
FlashOcc* [46]	C	R-50	43.5	38.9	14.7	56.0	58.2	30.2	22.0	23.9	16.6	42.8	46.4	75.3	39.4	45.0	51.1	64.4	70.7
M-CONet [33]	C+L	R-101	24.7	24.8	13.0	31.6	34.8	14.6	18.0	20.0	14.7	20.0	26.6	39.2	22.8	26.1	26.0	26.0	37.1
Co-Occ [26]	C+L	R-101	27.1	28.1	16.1	34.0	37.2	17.0	21.6	20.8	15.9	21.9	28.7	42.3	25.4	29.1	28.6	28.2	38.0
OccFusion [25]	C+L+R	R-101	27.3	27.1	**19.6**	33.7	36.2	21.7	24.8	25.3	16.3	21.8	30.0	39.5	19.9	24.9	26.5	28.9	40.4
DAOcc [44]	C+L	R-50	30.5	30.8	19.5	34.0	38.8	25.0	27.7	29.9	**22.5**	23.2	31.6	41.0	25.9	29.4	29.9	35.2	43.5
HGS-OCC	C	R-50	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 3. 3D Occupancy prediction performance on the SurroundOcc validation set. The best scores for each class are highlighted in bold. In the Input, the C, L, and R denote camera, LiDAR, and radar, respectively. * means the performance is achieved by our implementation using its official code.

ployed for BEV augmentation.

4.3. Comparing with SOTA methods

Occ3D-nuScenes. As shown in Tab.1, we present a quantitative comparison of existing state-of-the-art methods for 3D occupancy prediction tasks on Occ3D-nuScenes dataset. Our method, which employs a lightweight ResNet-50 backbone, achieves state-of-the-art performance in both mIoU and IoU across nearly half of the categories, , while also delivering the fastest inference speed. Notably, it outperforms even multi-modal fusion approaches, effectively balancing performance and speed to meet the real-time processing requirements of autonomous driving applications.

Moreover, we also conducted a Ray-IoU (proposed by [22]) performance comparison without visual masks, as Ray-IoU is chosen for its accuracy in evaluating occupancy predictions and addressing depth consistency issues in traditional voxel-based IoU metrics. Despite many existing image-based methods utilizing multi-frame fusion, as shown in Tab.2, HGS-Occ outperforms them both in terms of performance and speed, while using only ResNet-50 as the image backbone network. We also present visualizations on the Occ3D-nuScenes validation set for both day and night occlusion scenarios, as shown in Fig.6. Compared to the baseline method, our approach demonstrates superior performance in accurately identifying vehicles largely occluded by other vehicles and pedestrians that are partially occluded.

Surround-nuScenes. As shown in Tab.3, we provide a quantitative comparison on the Surroundocc validation set, highlighting the performance of our method, HGS-OCC, compared to other approaches. This success is attributed to the modeling of hybrid depth optimization and the fusion of multi-representation features. Notably, our method uses only a lightweight ResNet50 backbone and a lower resolution of 256×704, underscoring its effectiveness and efficiency.

Baseline	HDO	MGF point	MGF BEV	MGF voxel	iou(%)	mIoU(%)	Latency (s)
✓					90.27	37.84	-
✓	✓				93.05	44.60	-
✓	✓		✓		94.01	47.49	-
✓	✓	✓	✓		94.04	47.88	-
✓	✓		✓	✓	94.13	48.17	-
✓	✓	✓	✓	✓	94.17	48.34	-

Table 4. Ablation study on the Occ3D-nuScenes dataset. HDO: Hybrid Depth Optimization. MGF: Multi-Representation Geometric-Semantic Fusion.

D	l	IoU(%)	mIoU (%)
0.5	2	94.12	47.86
0.5	4	94.09	48.06
0.5	8	94.14	48.05
1	2	94.16	47.92
1	4	**94.17**	**48.34**
1	8	94.10	48.28

Table 5. Ablation study of the hyperparameter used in Hybrid Depth Optimization module on Occ3D-nuScenes.

4.4. Ablation study

The Effectiveness of Each Component. The results are shown in Tab.4 , we can observe that all components make their own performance contributions. The baseline achieves 90.27% IoU and 37.84% mIoU. Integrating Hybrid Depth Optimization (HDO) into the baseline results in improvements of 2.78% in IoU and 6.76% in mIoU. Further, multi-representational feature fusion is applied, with BEV fusion yielding the most significant performance gains. Although

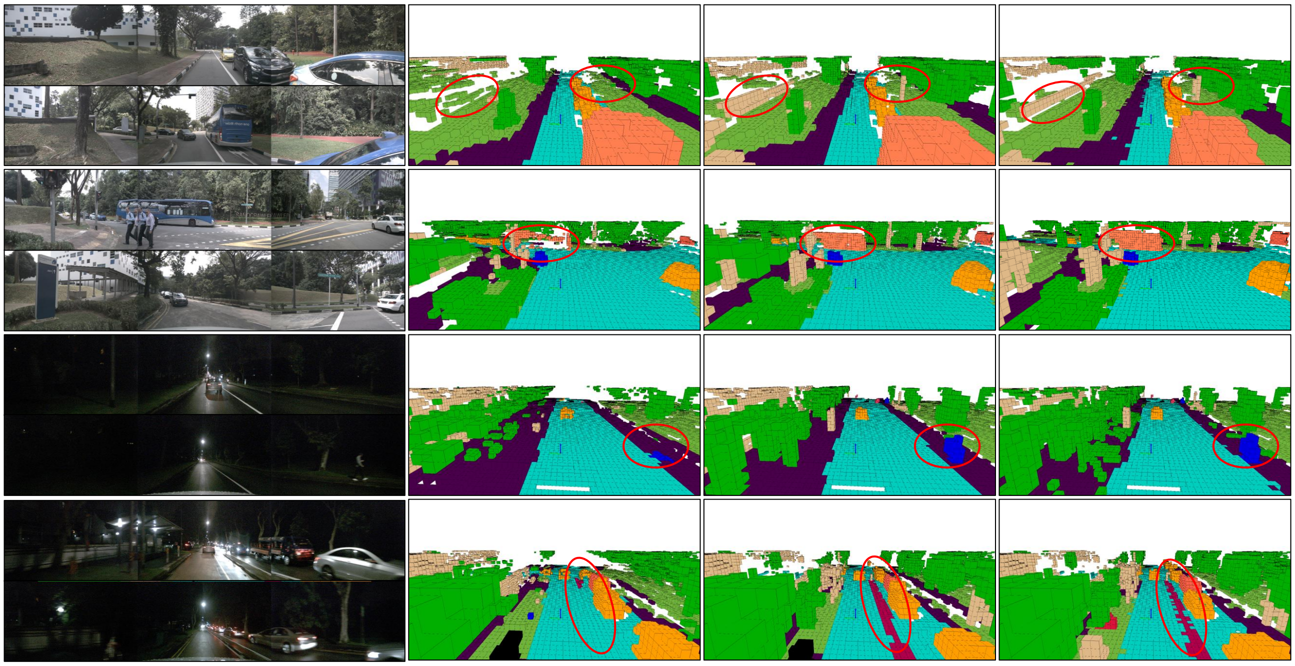


Figure 6. Qualitative results of SDG on the validation set of Occ3D-nuScenes. Each pair of rows displays results from day and low-light scene, respectively. Within each row, images from left to right represent the input images, baseline, our results, and the ground truth.

$Downsample$	IoU(%)	mIoU (%)	Latency (s)
4	94.12	47.86	-
8	94.09	48.06	-
16	94.14	48.05	-

Table 6. Ablation study on the impact of downsampling on performance and speed in Occ3D-nuScenes.

fusion of point cloud and voxel representations also enhances performance, the improvement is less pronounced due to their inherent sparsity. By combining hybrid depth optimization with multi-representation geometric-semantic fusion, we achieve a 3.9% improvement in IoU and a 10.5% increase in mIoU compared to the baseline.

The Effectiveness of Hybrid Depth Optimization. To further validate the effect of Hybrid Depth Optimization (HDO), we conducted hyperparameter analysis experiments. In this module, the range r of bilinear growth discretization and the number of diffusion feature layers l jointly control the generation of virtual points. As shown in the Tab.5, lower discretization range and fewer virtual points (e.g., $r = 0.5$ and $l = 2$) result in a slight performance decrease. However, excessively high discretization range and virtual points do not lead to additional performance gains, with optimal performance observed at $r = 1$ and $l = 4$.

Impact of downsampling on performance and speed. The downsampling factor influences depth map ambiguity, where pixel depths are averaged from neighboring pixels. Higher downsampling increases ambiguity but reduces computational load. An ablation study on the impact of downsampling on performance and speed is shown in Tab.6. The results show that 16x downsampling provides a small performance gain compared to lower downsampling factors, but with faster computation. 8x downsampling strikes a balance between performance and speed, while 4x downsampling yields limited performance improvement over 8x, at the cost of significantly slower computation.

5. Conclusion

In this paper, we introduce a fast 3D semantic occupancy prediction framework, termed HGS-OCC, designed to achieve higher accuracy and competitive inference speed. To generate virtual point seeds that preserve the 3D geometric structure, we propose a Hybrid Depth Optimization (HDO) module. This approach combines the relative depth from a Depth Angthing model fine-tuned with the absolute depth information provided by LiDAR for 3D modelling, effectively preserving the integrity of the 3D geometry while reducing the number of virtual point seeds. Based on this, we introduce a multi-representational geometric-semantic fusion module that utilizes efficient sparse 3D submanifold convolutions to generate geometric point cloud, BEV, and voxel features, which are then fused with the semantic features of the image. Our method achieves state-of-the-art performance and speed on the Occ3D-nuScenes and Surround-nuScenes datasets, demonstrating its effectiveness.

References

[1] Anh-Quan Cao and Raoul De Charette. Monoscene: Monocular 3d semantic scene completion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3991–4001, 2022. 1, 2

[2] Wanshui Gan, Ningkai Mo, Hongbin Xu, and Naoto Yokoya. A comprehensive framework for 3d occupancy estimation in autonomous driving. *IEEE Transactions on Intelligent Vehicles*, 2024. 6

[3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 3

[4] Jiawei Hou, Xiaoyan Li, Wenhao Guan, Gang Zhang, Di Feng, Yuheng Du, Xiangyang Xue, and Jian Pu. Fastocc: Accelerating 3d occupancy prediction by fusing the 2d bird's-eye view and perspective view. *arXiv preprint arXiv:2403.02710*, 2024. 3, 6

[5] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17853–17862, 2023. 1

[6] Junjie Huang and Guan Huang. Bevdet4d: Exploit temporal cues in multi-camera 3d object detection. *arXiv preprint arXiv:2203.17054*, 2022. 2, 6

[7] Junjie Huang, Guan Huang, Zheng Zhu, Yun Ye, and Dalong Du. Bevdet: High-performance multi-camera 3d object detection in bird-eye-view. *arXiv preprint arXiv:2112.11790*, 2021. 6

[8] Yuanhui Huang, Wenzhao Zheng, Yunpeng Zhang, Jie Zhou, and Jiwen Lu. Tri-perspective view for vision-based 3d semantic occupancy prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9223–9232, 2023. 2, 6

[9] Xiaosong Jia, Yulu Gao, Li Chen, Junchi Yan, Patrick Langechuan Liu, and Hongyang Li. Driveadapter: Breaking the coupling barrier of perception and planning in end-to-end autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 7953–7963, 2023. 1

[10] Yanqin Jiang, Li Zhang, Zhenwei Miao, Xiatian Zhu, Jin Gao, Weiming Hu, and Yu-Gang Jiang. Polarformer: Multi-camera 3d object detection with polar transformer. In *Proceedings of the AAAI conference on Artificial Intelligence*, pages 1042–1050, 2023. 2

[11] Xin Lai, Yukang Chen, Fanbin Lu, Jianhui Liu, and Jiaya Jia. Spherical transformer for lidar-based 3d recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17545–17555, 2023. 1

[12] Hongyang Li, Chonghao Sima, Jifeng Dai, Wenhai Wang, Lewei Lu, Huijie Wang, Jia Zeng, Zhiqi Li, Jiazhi Yang, Hanming Deng, et al. Delving into the devils of bird's-eye-view perception: A review, evaluation and recipe. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023. 2

[13] Jiale Li, Hang Dai, Hao Han, and Yong Ding. Mseg3d: Multi-modal 3d semantic segmentation for autonomous driving. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 21694–21704, 2023. 1

[14] Yinhao Li, Han Bao, Zheng Ge, Jinrong Yang, Jianjian Sun, and Zeming Li. Bevstereo: Enhancing depth estimation in multi-view 3d object detection with temporal stereo. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1486–1494, 2023. 1

[15] Yinhao Li, Zheng Ge, Guanyi Yu, Jinrong Yang, Zengran Wang, Yukang Shi, Jianjian Sun, and Zeming Li. Bevdepth: Acquisition of reliable depth for multi-view 3d object detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1477–1485, 2023. 2

[16] Yiming Li, Zhiding Yu, Christopher Choy, Chaowei Xiao, Jose M Alvarez, Sanja Fidler, Chen Feng, and Anima Anandkumar. Voxformer: Sparse voxel transformer for camera-based 3d semantic scene completion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9087–9098, 2023. 1, 3

[17] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. Bevformer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In *European conference on computer vision*, pages 1–18. Springer, 2022. 1, 6

[18] Zhiqi Li, Zhiding Yu, David Austin, Mingsheng Fang, Shiyi Lan, Jan Kautz, and Jose M Alvarez. Fb-occ: 3d occupancy prediction based on forward-backward view transformation. *arXiv preprint arXiv:2307.01492*, 2023. 3, 6

[19] Tingting Liang, Hongwei Xie, Kaicheng Yu, Zhongyu Xia, Zhiwei Lin, Yongtao Wang, Tao Tang, Bing Wang, and Zhi Tang. Bevfusion: A simple and robust lidar-camera fusion framework. *Advances in Neural Information Processing Systems*, 35:10421–10434, 2022. 2

[20] Tingting Liang, Hongwei Xie, Kaicheng Yu, Zhongyu Xia, Zhiwei Lin, Yongtao Wang, Tao Tang, Bing Wang, and Zhi Tang. Bevfusion: A simple and robust lidar-camera fusion framework. *Advances in Neural Information Processing Systems*, 35:10421–10434, 2022. 1

[21] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2117–2125, 2017. 3

[22] Haisong Liu, Haiguang Wang, Yang Chen, Zetong Yang, Jia Zeng, Li Chen, and Limin Wang. Fully sparse 3d panoptic occupancy prediction. *arXiv preprint arXiv:2312.17118*, 2023. 6, 7

[23] Zhijian Liu, Haotian Tang, Alexander Amini, Xinyu Yang, Huizi Mao, Daniela L Rus, and Song Han. Bevfusion: Multi-task multi-sensor fusion with unified bird's-eye view representation. In *2023 IEEE international conference on robotics and automation (ICRA)*, pages 2774–2781. IEEE, 2023. 1

[24] Qihang Ma, Xin Tan, Yanyun Qu, Lizhuang Ma, Zhizhong Zhang, and Yuan Xie. Cotr: Compact occupancy transformer

for vision-based 3d occupancy prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19936–19945, 2024. 1, 3, 6

[25] Zhenxing Ming, Julie Stephany Berrio, Mao Shan, and Stewart Worrall. Occfusion: Multi-sensor fusion framework for 3d semantic occupancy prediction. *IEEE Transactions on Intelligent Vehicles*, 2024. 6, 7

[26] Jingyi Pan, Zipeng Wang, and Lin Wang. Co-occ: Coupling explicit feature fusion with volume rendering regularization for multi-modal 3d semantic occupancy prediction. *IEEE Robotics and Automation Letters*, 2024. 7

[27] Mingjie Pan, Jiaming Liu, Renrui Zhang, Peixiang Huang, Xiaoqi Li, Li Liu, and Shanghang Zhang. Renderocc: Vision-centric 3d occupancy prediction with 2d rendering supervision. *arXiv preprint arXiv:2309.09502*, 2023. 3, 6

[28] Mingjie Pan, Li Liu, Jiaming Liu, Peixiang Huang, Longlong Wang, Shanghang Zhang, Shaoqing Xu, Zhiyi Lai, and Kuiyuan Yang. Uniocc: Unifying vision-centric 3d occupancy prediction with geometric and semantic rendering. *arXiv preprint arXiv:2306.09117*, 2023. 3

[29] Jonah Philion and Sanja Fidler. Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XIV 16*, pages 194–210. Springer, 2020. 2

[30] Cody Reading, Ali Harakeh, Julia Chae, and Steven L Waslander. Categorical depth distribution network for monocular 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8555–8564, 2021. 2, 4

[31] Haotian Tang, Zhijian Liu, Shengyu Zhao, Yujun Lin, Ji Lin, Hanrui Wang, and Song Han. Searching efficient 3d architectures with sparse point-voxel convolution. In *European conference on computer vision*, pages 685–702. Springer, 2020. 4, 6

[32] Xiaoyu Tian, Tao Jiang, Longfei Yun, Yucheng Mao, Huitong Yang, Yue Wang, Yilun Wang, and Hang Zhao. Occ3d: A large-scale 3d occupancy prediction benchmark for autonomous driving. *Advances in Neural Information Processing Systems*, 36, 2024. 1, 6

[33] Xiaofeng Wang, Zheng Zhu, Wenbo Xu, Yunpeng Zhang, Yi Wei, Xu Chi, Yun Ye, Dalong Du, Jiwen Lu, and Xingang Wang. Openoccupancy: A large scale benchmark for surrounding semantic occupancy perception. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 17850–17859, 2023. 1, 7

[34] Yue Wang, Vitor Campagnolo Guizilini, Tianyuan Zhang, Yilun Wang, Hang Zhao, and Justin Solomon. Detr3d: 3d object detection from multi-view images via 3d-to-2d queries. In *Conference on Robot Learning*, pages 180–191. PMLR, 2022. 2

[35] Yuqi Wang, Yuntao Chen, and Zhaoxiang Zhang. Frustumformer: Adaptive instance-aware resampling for multi-view 3d detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5096–5105, 2023. 2

[36] Yuqi Wang, Yuntao Chen, Xingyu Liao, Lue Fan, and Zhaoxiang Zhang. Panoocc: Unified occupancy representation for camera-based 3d panoptic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 17158–17168, 2024. 1, 6

[37] Zirui Wang, Shangzhe Wu, Weidi Xie, Min Chen, and Victor Adrian Prisacariu. Nerf–: Neural radiance fields without known camera parameters. *arXiv preprint arXiv:2102.07064*, 2021. 3

[38] Yi Wei, Linqing Zhao, Wenzhao Zheng, Zheng Zhu, Jie Zhou, and Jiwen Lu. Surroundocc: Multi-camera 3d occupancy prediction for autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21729–21740, 2023. 3, 6, 7

[39] Philipp Wolters, Johannes Gilg, Torben Teepe, Fabian Herzog, Anouar Laouichi, Martin Hofmann, and Gerhard Rigoll. Unleashing hydra: Hybrid fusion, depth consistency and radar for unified 3d perception. *arXiv preprint arXiv:2403.07746*, 2024. 6

[40] Xiaoyang Wu, Li Jiang, Peng-Shuai Wang, Zhijian Liu, Xihui Liu, Yu Qiao, Wanli Ouyang, Tong He, and Hengshuang Zhao. Point transformer v3: Simpler faster stronger. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4840–4851, 2024. 1

[41] Yichen Xie, Chenfeng Xu, Marie-Julie Rakotosaona, Patrick Rim, Federico Tombari, Kurt Keutzer, Masayoshi Tomizuka, and Wei Zhan. Sparsefusion: Fusing multi-modal sparse representations for multi-sensor 3d object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 17591–17602, 2023. 1

[42] Xu Yan, Jiantao Gao, Chaoda Zheng, Chao Zheng, Ruimao Zhang, Shuguang Cui, and Zhen Li. 2dpass: 2d priors assisted semantic segmentation on lidar point clouds. In *European Conference on Computer Vision*, pages 677–695. Springer, 2022. 1

[43] Chenyu Yang, Yuntao Chen, Hao Tian, Chenxin Tao, Xizhou Zhu, Zhaoxiang Zhang, Gao Huang, Hongyang Li, Yu Qiao, Lewei Lu, et al. Bevformer v2: Adapting modern image backbones to bird's-eye-view recognition via perspective supervision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17830–17839, 2023. 2

[44] Zhen Yang, Yanpeng Dong, and Heng Wang. Daocc: 3d object detection assisted multi-sensor fusion for 3d occupancy prediction. *arXiv preprint arXiv:2409.19972*, 2024. 7

[45] Junbo Yin, Jianbing Shen, Runnan Chen, Wei Li, Ruigang Yang, Pascal Frossard, and Wenguan Wang. Is-fusion: Instance-scene collaborative fusion for multimodal 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14905–14915, 2024. 1

[46] Zichen Yu, Changyong Shu, Jiajun Deng, Kangjie Lu, Zongdai Liu, Jiangyong Yu, Dawei Yang, Hui Li, and Yan Chen. Flashocc: Fast and memory-efficient occupancy prediction via channel-to-height plugin. *arXiv preprint arXiv:2311.12058*, 2023. 3, 6, 7

[47] Zichen Yu, Changyong Shu, Qianpu Sun, Yifan Bian, Xiaobao Wei, Jiangyong Yu, Zongdai Liu, Dawei Yang, Hui Li, and Yan Chen. Panoptic-flashocc: An efficient baseline

to marry semantic occupancy with panoptic via instance center. *arXiv preprint arXiv:2406.10527*, 2024. 6

[48] Yunpeng Zhang, Zheng Zhu, and Dalong Du. Occformer: Dual-path transformer for vision-based 3d semantic occupancy prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9433–9443, 2023. 1, 6, 7

[49] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. *arXiv preprint arXiv:2010.04159*, 2020. 2

[50] Xinge Zhu, Hui Zhou, Tai Wang, Fangzhou Hong, Yuexin Ma, Wei Li, Hongsheng Li, and Dahua Lin. Cylindrical and asymmetrical 3d convolution networks for lidar segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9939–9948, 2021. 1